MegaFlash PCB Assembly Notes

PCB Revisions

Three PCB revisions have been developed. Rev 1.0a PCB is smaller and has fewer components. But it is very difficult to hand assemble.

So, a more DIY friendly Rev 1.1b was developed. Only through hole parts and 1.27mm pitch SOICs are used. Rev 1.2 replaces the GAL16V8 PLD with three 74HC138.

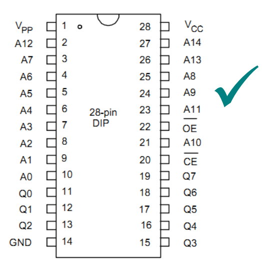
All three PCB revisions are functionally identical.

Part Selections and Alternatives

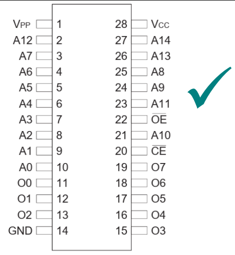
<u>ROM Chip</u>

A ROM chip replacement of Apple System ROM is needed for MegaFlash to function. The recommended part is Winbond W27E257 or W27C257 EEPROM. They are pin compatible with IIc/IIc+ ROM chip.

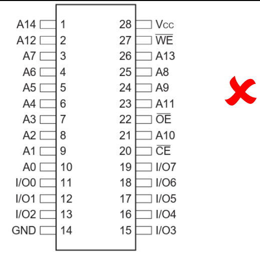
AT27C256R should also be pin compatible. But it is one-time programmable only. Note that 28C256 <u>is not</u> pin compatible.



W27E257/W27C257



AT27C256R



28C256

<u>Pico Controller Board</u>

As of July-2025, there are 4 models of Pico Controller Board − Pico, Pico W, Pico 2 and Pico 2W. The W variants support WIFI. All 4 Pico boards can be used to build MegaFlash. The firmware of Pico and Pico 2 are different. So, the correct firmware file must be installed.

Pico and Pico W board have 260kB of RAM only. So, the RAM Disk is limited to 140kB (vs 400kb on Pico 2) and Apple Memory Expansion card emulation is not supported if Pico controller board is used.

Pico 2W is recommended since it has WIFI and more RAM.

5V to 3.3V level shifter

On Rev 1.1b and 1.2 PCB, 74LVC8T245 Level shifter ICs are used. 75LVC4245A is not compatible with Rev 1.2 PCB. On Rev 1.2 PCB, the direction of the transceiver is flipped. The 5V signals from Apple are connected to Input B. 75LVC4245A only allows 5V signals from Input A.

Pull-up resistors

All the resistors on PCB Rev 1.1b are pull-up resistors. The bus transceiver IC 74LVC8T245 is a CMOS device. Its inputs should not be floating at all times. So, pull-up resistors are added. RN1, RN2 and RN3 are pull-up resistor for the bus transceiver input. Any value from 10kΩ to 100kΩ should do the job.

On Rev 1.2 PCB, the pull-up resistors are removed. It is found that the board functions normally without those resistors.

R1 and R2 are pull-up resistors for the flash chips. It pulls the chip select line high so that the chip is disabled if the /CS line is not driven.

On Rev 1.0a PCB, 74LVCH16T245 transceiver IC is used. This chip has a bus-hold function and does not require pull-up resistors. But the flash chips still require pull-up resistors.

Pin Header

The pin header specified in bill of material is from Samtec. It is gold plated but very expensive. The alternative is double row 2.54mm pitch 19mm length pin header, which can is available from eBay or AliExpress.

Schottky Diode

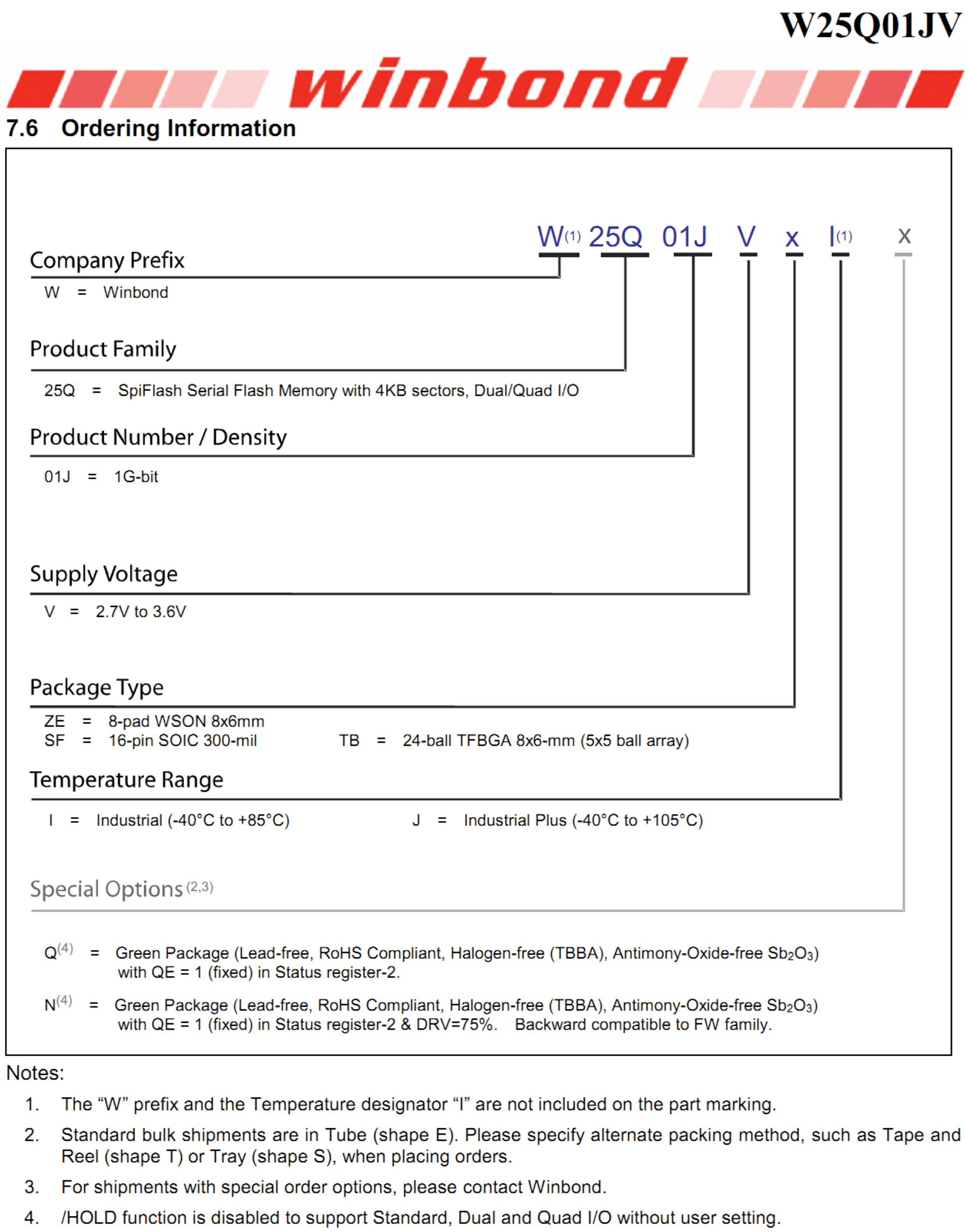
The diode is to prevent electrical current from flowing back to Apple if the USB port of Pico board is connected to a power source. On Rev 1.1b PCB, both through hole and SMD footprints are provided. Only one Schottky diode is needed.

Flash chips

One or two flash chips can be installed. If only one chip is installed, it must be placed to the position labeled Flash 1 (i.e. U5 on Rev 1.0a and Rev1.1b PCB). The following flash chips are supported by the software.

Manufacturer	Part Number	Capacity	JEDEC ID	Tested
Winbond	W25Q01JV	128MB	EF4021h	Yes
Winbond	W25Q01JV-DTR	128MB	EF7021h	Yes
Winbond	W25Q512JV	64MB	EF4020h	No
Winbond	W25Q512JV-DTR	64MB	EF7020h	No
Winbond	W25Q02JV-DTR	256MB	EF7022h	No
Alliance	AS25F3512MQ	64MB	204020h	Yes

Here is the meaning of the part number. The last letter can be Q, M or N. All three variants should be compatible.



W25Q01JV

winbond

7.6 Ordering Information

W(1) 25Q 01J V x I(1) x

Company Prefix

W = Winbond

Product Family

25Q = SpiFlash Serial Flash Memory with 4KB sectors, Dual/Quad I/O

Product Number / Density

01J = 1G-bit

Supply Voltage

V = 2.7V to 3.6V

Package Type

ZE = 8-pad WSON 8x6mm
SF = 16-pin SOIC 300-mil
TB = 24-ball TFBGA 8x6-mm (5x5 ball array)

Temperature Range

I = Industrial (-40°C to +85°C)
J = Industrial Plus (-40°C to +105°C)

Special Options (2,3)

Q(4) = Green Package (Lead-free, RoHS Compliant, Halogen-free (TBBA), Antimony-Oxide-free Sb_2O_3) with QE = 1 (fixed) in Status register-2.

N(4) = Green Package (Lead-free, RoHS Compliant, Halogen-free (TBBA), Antimony-Oxide-free Sb_2O_3) with QE = 1 (fixed) in Status register-2 & DRV=75%. Backward compatible to FW family.

Notes:

1. The “W” prefix and the Temperature designator “I” are not included on the part marking.
2. Standard bulk shipments are in Tube (shape E). Please specify alternate packing method, such as Tape and Reel (shape T) or Tray (shape S), when placing orders.
3. For shipments with special order options, please contact Winbond.
4. /HOLD function is disabled to support Standard, Dual and Quad I/O without user setting.

Note:

1) The total capacity of the two flash chips is limited to 256MB.
2) –DTR variant stands for Double Transfer Rate. The SPI interface of Pico does not support double transfer rate. But the chips can be operated in single transfer rate mode. So, they are compatible.
3) The software should recognize W25Q512JV and W25Q02JV. But they are not tested.

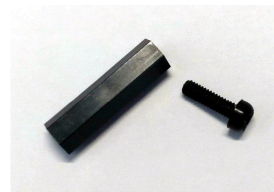
Construction Tips

Clearance

The clearance between MegaFlash PCB and bottom of the keyboard is very tight. To avoid short-circuit, cut off the component leads as much as possible.

It is recommended to cover the solder joints with electrical tape, kapton tape or simply a piece of paper.

Nylon Standoff



The M2.5 mounting hole on Rev 1.1b PCB is for mounting a nylon standoff to support the weight of the PCB since it is wider and heavier than Rev 1.0a PCB. The diameter of the hole is 2.5mm. The recommended length of the standoff is 18mm.

Other Information

A2 and A3 Address Lines

GPIO 8 and 9 of Pico board are reserved for Apple address line A2 and A3. Since the memory expansion connector does not provide those address lines. On Rev 1.0 PCB, those lines are not connected. On Rev 1.0a, 1.1b and 1.2 PCB, those lines are connected to ground.

3.3V Regulator

On Rev 1.0 PCB, there is a 3.3V voltage regulator to provide 3.3V power to the transceiver IC. The regulator is removed on Rev 1.0a and Rev 1.1b and the transceiver ICs are powered by the regulator on the Pico controller board.